

Midterm Exam**10/11/2022****Problem 1 part 1)**

Given the image region as shown in Figure 1 and $V = \{1\}$, what is the shortest m-path between p (the pixel at the upper-left corner) and q (the pixel at the bottom-right corner)? Give the length of the path if the path exists. (15 points)

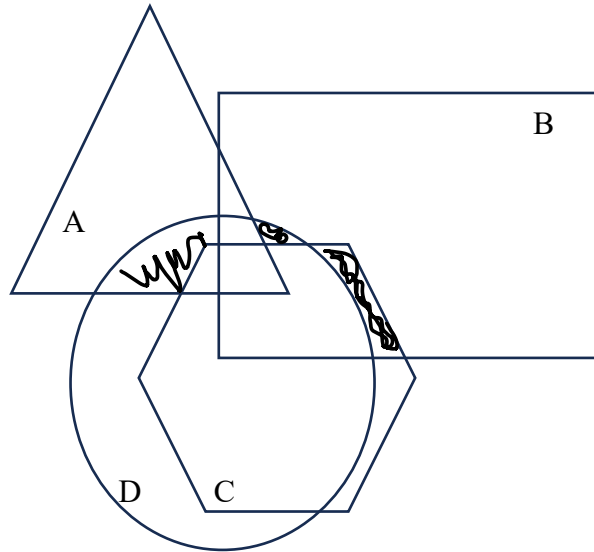
1	0	1	1	0
0	1	0	1	0
0	0	1	1	0
1	1	0	1	0
0	1	1	1	1

Problem 1 part 2)

To take a picture for fireworks at night, give at least two operations that should be applied to improve the visualization quality of the image and explain. (15 points)

Problem 2)

A, B, C and D are four sets. Give expressions for the set shown shaded in Figure 2 in terms of A, B, C, and D. Note that all of the shaded areas constitute of one set



Problem 3)

Given a 3x3 spatial filter $\frac{1}{9} \begin{bmatrix} -1 & -1 & -1 \\ -1 & -17 & -1 \\ -1 & -1 & -1 \end{bmatrix}$ answer the following questions:

- a) What type of filter is it? When applying this filter on an image, what kind of effect you will expect in the output image? Why? (Hint is this filter an image smoothing filter, an image sharpening filter or an edge detector?) (10 points)

- b) Perform the image convolution using this filter on a 3x3 image $\begin{bmatrix} 3 & 5 & 12 \\ 7 & 18 & 11 \\ 5 & 9 & 15 \end{bmatrix}$. Give the

CROPPED filtering result for the output image. (Hint: you need to assume an appropriate zero mapping) (15 points)

Problem 4)

An image with intensities in the range $r \in [0, L - 1]$ has the pdf

$$p_r(r) = \begin{cases} \frac{4r}{3(L-1)^2} & 0 \leq r \leq L-1 \\ 0 & \text{otherwise} \end{cases}, \text{ where } L-1 \text{ is the maximum intensity value. It is desired to}$$

transform the intensity levels of this image so that they will have the specified pdf

$$p_z(z) = \begin{cases} \frac{2z}{(L-1)^2} & 0 \leq z \leq L-1 \\ 0 & \text{otherwise} \end{cases}. \text{ Assume that the intensity value is continuous and find the}$$

transformation function $z = T(r)$ in terms of r and z that will accomplish this. (25 points)

(Hints: The transformation function of histogram equalization is given as

$$s = T(r) = (L-1) \int_0^r p_r(w) dw)$$

Bonus Question

Perform the Fourier transform to a function as shown in Figure 3. (20 points)

(Hint: the definition of the Fourier transform of a function is $F(u) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi ut} dt$)

